

		Root mean square speed = $\sqrt{\frac{3}{2} \times 0.004}$
13	B	$pV = nRT = \frac{m}{M}RT$ At constant pressure and volume, $mT = \text{constant}$. $m_2 T_2 = m_1 T_1$ $\textcircled{R} \quad m_2 = \frac{m_1 T_1}{T_2} = \frac{(273+10)}{(273+30)} \times 15 = 14 \text{ kg}$
14	B	$Q = Pt = mc\Delta t$